

SECTION 15961-A
PIPE AND DUCT HANGERS, SUPPORTS,
ANCHORS, GUIDES, AND SEALS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions, apply to this Section.

1.2 SUMMARY

- A. Section describes pipe hangers, supports, anchors, guides, and seals required for mechanical systems installations. Pipe hanger material shall conform to the minimum level of acceptance set forth in Manufacturers Standardization Society (MSS) SP-58. This specification indicates general MSS types only and manufacturers' other standard products are acceptable under Standard SP-58, if they meet dimensional and load rating limitation. Hangers, supports, anchors, guides, and restraints shall be selected to withstand all static and dynamic condition of loading to which piping and associated equipment may be subjected. Where supplementary structural members are designed, they shall be designed by the Contractor for the specific loads they are to support in accordance with the American Institute of Steel Construction (AISC) Specification For the Design, Fabrication, and Erection of Structural Steel for Buildings. Allowable stress levels listed in Table 2 of MSS SP-58 shall be used in Contractor's application of hanger assemblies.

1.3 REFERENCES AND APPROVALS

- A. ANSI B 31.1, Power Piping.
- B. ANSI B31.9, Building Services Piping.
- C. MSS SP-58, Pipe Hangers and Supports - Materials, Design, and Manufacturing.
- D. MSS SP-69, Pipe hanger and Supports Selection and Application.
- E. MSS SP-89, Pipe Hangers and Supports Fabrication and Installation Practices.

- F. MSS SP-90, Guidelines on Terminology for Pipe Hangers and Supports.
- G. AWS D1.1, Structural Welding Code.
- H. SMACNA's relevant standards.

PART 2 - PRODUCTS

2.1 INSERTS, EXPANSION SHIELDS, FASTENERS, AND CLAMPS

- A. Inserts for use in poured-in-place concrete shall have malleable iron cases or galvanized steel shells and expander plugs for threaded connection with lateral adjustment, top slot for reinforcing rods and lugs for attaching to forms. Provide manufactured threaded rod inserts for metal decks with concrete, retaining rod to extend 2 inches minimum above top of deck flutes. Inserts may not be used on roof decks without written specific acceptance by the Construction Manager.
- B. Size inserts to fit threaded hanger rods.
- C. Provide ICBO and ASTM E-488 tested drilled insert expansion shields, inserts, and fasteners on concrete walls and floors and sides of concrete beams. Powder-driven fasteners, expansion nail type anchors, or friction spring-type clips will not be permitted. Fastener devices exposed to corrosive fluids shall be of 316 stainless steel material.
- D. In areas of metal decking, pipe and duct shall be supported from beams or joists and intermediate steel with UL/FM MSS Type 23 malleable iron "C" clamps with Grinnell Figure No. 89 retainer straps or equal. Provide angles or channels to span between joists or beams as required to maintain support centers specified herein.

2.2 PIPE HANGERS AND SUPPORTS

- A. Hangers for Uninsulated Pipe: An adjustable wrought steel clevis MSS Type-1.
- B. Hangers for Cold Insulated Pipe, All Sizes: An adjustable wrought steel clevis, MSS Type-1, sized to suit 360 degrees high-density insulation inserts.

- C. Hangers for Hot Insulated Pipe, Sizes 1/2 inch to 1-1/2 inch: An adjustable wrought steel clevis, hanger, MSS Type-1, sized to suit a 180 degrees, with MSS Type-40, galvanized insulation protection saddle.
- D. Hangers for Hot Insulated Steel Pipe, Sizes 2 inches and Larger: Adjustable roll hanger with MSS Type-43 steel yoke and cast iron roller, or Type-3 double bolt pipe clamp as required for application.
- E. Hangers for Hot Insulated Copper Pipe, Sizes 2 inches and Larger: Same as for steel except with 360 degrees high-density insert.
- F. Support devices for all plastic pipes should not cause point loading stresses on the piping. Pipe should be supported by means of clips which do not fix the pipe in axial direction, such as Clic supports for 2 inches and under pipe sizes and Agru/Asahi polypropylene composite hanger for 2 inch and larger pipe sizes.
- G. Provide roller MSS Types-41, 43, 44, or slide Type MSS-35 at significant movement location to permit free movement of piping where expansion and contraction of systems piping will occur.
- H. Insulated pipe supports shall be utilized on insulated pipe lines. Proper models to be selected by hanger load, hanger style, and permissible hanger span distances specified herein.
 - 1. Model designations are by Pipe Shields, Inc., or approved equal. Equals must submit documentation demonstrating compliance with applicable MSS and ANSI standards.
 - 2. Models:
 - a. Insulated Pipe Supports:
 - 1) Pipe Supported on Rod Hangers: Use Models A1000, A2000, A3000, A4000, and A9000.
 - 2) Pipe Supported on Flat Surfaces: Use Models A1000, A2000, A5000, A6000, and A7000 to A7400 Series.
 - 3) Pipe Supported on Pipe Rolls: Use Models A3000, A4000, A5000, A6000, and A8000 to A8400 Series.
 - 4) Pipe Risers: Models E1000 to E1300 Series and E2000 to E2300 Series.
 - 3. Seismic pipe supports locations shall require the following:
 - a. Properly selected Pipe Shields, Inc., A9000 Series at all seismic support contact points to

- meet the seismic design specified and document with certified layout chart.
- b. Certified load capacity sheets, reviewed by registered engineer.
- I. Trapeze: Channel ASTM A 570, Grade 33, Hot Rolled Structural Quality; finish, ASTM B-633.
- J. Channels: Located in clean spaces shall be painted.
- K. Wall Support for Pipe, Sizes 4 inches and Larger: Medium welded steel bracket, MSS Type-32. Finish; ASTM A-153; B-633.
- L. Vertical Support: Steel and copper pipe riser clamps, MSS Types-8 or -42 as applicable to service.
- M. Rod Couplings: Grinnell straight Figure 136, UL/FM approved or equal.
- N. Floor Support for Hot and Cold Pipes: Adjustable cast iron roll and stand with concrete pier or steel support, MSS Type-38.
- O. Center Beam Clamps: MSS Types -21 or -28 as required by upper attachment application.
- P. Install supports to impede disengagement of hanger, by movement of supported pipe.
- Q. Provide copper-plated hangers, Grinnell Figure CT-65 for copper piping or Grinnell Figure CT-169 between hanger or support and piping.
- R. Provide angles or channels to span joists and distribute load.
- S. Welded Beam Attachments: MSS Types-22 or -57 as required to suit hanger application.
- T. Pipe Guides: Grinnell Figure 255 spider guide or equal.
- U. Steel Pipe Slides: Grinnell Figure 257 or equal, MSS Type-35.
- V. Do not use wire for either temporary or permanent hanger or support purposes.

2.3 PIPE ISOLATORS

- A. Provide each hanger or clamp for uninsulated piping with a metal backed pipe isolating material to isolate sound,

vibration and electrolysis. Isolators are not required for fire protection, sprinkler piping, waste, vent, gas, and downspout piping.

2.4 HANGER RODS

- A. Provide steel hanger rods, appropriately threaded. Provide connection points with jamb nuts or double nuts. For highly corrosive environment FRP hanger rods employing vinyl ester resin shall be used.

2.5 DUCT HANGERS AND SUPPORTS

- A. Per latest SMACNA's Standards.
- B. Provide galvanized steel angles, channels, stays, rods, etc., for duct support and reinforcement unless painted where specified.

2.6 FLASHING

- A. Steel Flashing: 26 gauge galvanized steel.
- B. Lead Flashing: 4 lb/sq ft sheet lead for waterproofing; 1 lb/sq ft sheet lead for soundproofing.
- C. Safes: 5 lb/sq ft sheet lead or 8 mil thick neoprene.
- D. Caps: Steel, 22 gauge minimum, 16 gauge at fire-resistant structures.

2.7 SLEEVES

- A. Sleeves for Pipes Passing Through Nonrated, Nonbearing Walls and Floors: 20 gauge, galvanized sheet metal tube with welded longitudinal joints.
- B. Sleeves for Pipes Passing Through Beams and Footings: Schedule 40 black steel pipe. For waterproof sleeves, use Thunderline Link-Seal, Calpico Sealing Linx or equal.
- C. Penetration through Fire-Rated Walls and Floor:
 - 1. Manufactured, pre-engineered fire-stop systems shall be installed at all penetrations of fire and smoke rated partitions and decks. Fire-rated assemblies will display a Listing Label or Stamp by UL or FM, documenting passage of test and code requirements as stipulated in ASTM E 814, E 119; UL 1479, and UBC 43-6.
 - 2. Installed assemblies shall have the same ratings as the fire separation in which they are installed.

3. Openings with sleeves cast-in-place in masonry do not require fasteners into the surface plate.
 4. Fire-stop systems shall be installed in the fire separation wall, floor or roof, to conform to the design listing.
- D. Open voids and cavities occurring in pipe sleeves passing through rated walls and floors shall be completely sealed with UL classified penetration sealant installed in strict accordance with the manufacturer's recommendations.
 - E. Sleeves for Ducts: Form with galvanized steel.
 - F. Size sleeves large enough to allow for movement due to expansion and to provide for continuous insulation.

2.8 SEISMIC EXPANSION COMPENSATORS

- A. Compensators for Hydronic Piping Systems 3 inches and Larger: Hyspan Series 1500 Universal with flanged ends, tie rods, and 321 stainless steel laminated bellows or equal.
- B. Compensators for Hydronic Piping 2-1/2 inches and Smaller: Hyspan 321 stainless steel braided hose or equal. Provide Series 4505 with threaded ends or Series 4510 with sweat ends.
- C. Provide units suitable for minimum 10.3 barg at 204 degrees C.
- D. Units shall be manufactured and installed in accordance with Expansion Joint Manufacturers Association (EJMA) recommendations and shall provide for multiple plane movement for seismic isolation.

2.9 PENETRATION SEALANTS

- A. General: Provide manufacturer's standard barrier sheets and fire-stopping sealant, with accessory materials, having fire-resistance ratings indicated as established by testing identical assemblies per ASTM E 814 by Underwriters Laboratories, Inc.
 1. Fire Barrier Sheets: 3M Fire Barrier Composite Sheet System CS-195 and FS-195 wrap/strip with associated accessories or equal.
 2. Fire-Stopping Sealant: 3M Fire Barrier Caulk CP 25, CP 25WB, CP25S/L, CP25N/S, Putty 303 or equal.

PART 3 - EXECUTION

3.1 INSERTS AND EXPANSION SHIELDS

- A. Provide MSS Type-18 or equal inserts for suspending hangers from reinforced concrete slabs and sides of reinforced concrete beams where continuous embedded insert type channel has not be provided.
- B. Set inserts in position in advance of concrete work. Provide reinforcement rod in concrete for inserts carrying pipe over 4 inches in diameter or ducts over 60 inches wide.
- C. Where concrete slabs form finished ceiling, finish inserts flush with slab surface.
- D. Locate expansion shields in concrete beams a minimum of 6 inches above bottom of beam.
- E. Do not use power-driven fasteners or friction spring-type clips.

3.2 PIPE HANGERS AND SUPPORTS

- A. Support horizontal steel and copper piping per B31.1 and MSS SP-58, as follows:

Pipe Size	Steel (m)	Copper (m)	Hanger Rod Diameter (mm)
1/4 - 3/8	1.2	1.2	10
1/2 - 1- 1/2	2.1	1.5	10
2 - 4	3.0	2.4	14
6 - 12	5.2	4.3	16
14 - 18	7.6		24
20 and up	*		*

*Supply certified calculations

- B. Support spacing for plastic pipes will follow the manufacturer's recommendations.
- C. Common trapeze cross member supports are preferred over individual hangers where there are multiple pipes. The use of a common trapeze cross member to hang multiple piping systems is permissible with the following conditions:

1. Cross members will be sized to support the composite load of all pipes.
 2. Pipes will be hung on top side of cross members only, no inverted lines.
 3. Width of cross members to be sufficient to secure full width base of pipe guides, roller chairs, pipe roll stand, pipe slide assemblies, etc.
 4. Pipes to be retained in place by MSS Type-24 U-Bolt or Type-26 bolted clamp.
 5. Cross members shall have a support rod located at each end of member. Cantilevered structural channel members to support pipes, is not acceptable.
 6. Trapezes assemblies shall be seismic braced in accordance with Section 15245.
 7. Pipes subject to thermal expansion shall be free to roll axially or slide.
 8. Structural type tubes shall be used for large bore pipe and common members when composite pipe loads exceed structural channel design capability. Structural tube members layout and fabrication to be field determined to suit individual system applications.
- D. Install hangers to provide minimum 38 mm clear space between finished covering and adjacent work.
- E. Random hangers shall be provided at each change in direction, at ends of branches, at base and top of risers, pipes and drops, and wherever necessary to prevent sag, bending, or vibration, in addition to the above listed hanger spacing. Provide MSS Type hangers.
- F. Use hangers which are vertically adjustable 38 mm inches minimum after piping is erected.
- G. Support horizontal cast iron soil and vent pipe near each joint, with MSS Type-1, Grinnell Figure 590 or equal, at 1.5 m maximum spacing between hangers and with no more than 2 joints occurring between any 2 hangers. Cast iron piping joined with a shield and clamp assembly, where the top of pipe is 305 mm or more from the supporting structure, shall be braced on each side of a change in direction of 90 degrees or more.
- H. Support steel and copper vertical piping at each floor. Support vertical cast iron pipe at each floor level with MSS Type-8 clamp. Provide joint stabilizers at cast iron pipe joints in between floors.
- I. Where practical, support riser piping independently of connected horizontal piping.

- J. Support nonmetallic piping with a sufficient number of hangers to prevent sagging and misalignment.
- K. All minor components such as straps, threaded rods, nuts, and washers shall be galvanized or cadmium plated. All other components, including angle iron, channels, and Unistrut, shall be factory or field painted. For applications in highly corrosive areas, FRP structural components and other parts in suitable plastic materials shall be provided.
- L. Anchors, guides, and other restraints shall be provided to secure the piping at their respective locations against movement in indicated planes or directions, while permitting free movement elsewhere. They shall be structurally suitable to withstand the thrusts, moments, and other imposed loads.
- M. Required Guides: Where bellows or slip-type expansion joints are used, appropriate anchors and guides shall be provided to direct expansion movement along the axis of the joint, in accordance with manufacturer's recommendations and installation instructions.
- N. Pipe hanger plan location with offset has been indicated for systems with anticipated thermal growth at support points.
- O. Piping and equipment anchors, stands, brackets, kneebraces, supplementary steel and stanchions, are to be field constructed and erected to withstand required design loads and seismic loads. Refer to Section 15245.
- P. Support piping with vibration isolators as specified in Section 15240.

3.3 DUCT HANGERS AND SUPPORTS

- A. Duct hangers, supports, and installation per latest SMACNA's Standards. All minor components such as straps, threaded rods, nuts, and washers shall be galvanized or cadmium plated.
- B. Specific details shown on Drawings take precedence over latest SMACNA's requirements.
- C. Support ductwork with vibration isolators as specified in Section 15240.

3.4 PRIMING

- A. Prime coat exposed steel hangers and supports which are not galvanized, cadmium plated, or factory painted. Hangers and supports located in crawl spaces, pipes shafts and suspended ceiling spaces are not considered exposed.

3.5 FLASHING

- A. Flash and counterflash where mechanical equipment passes through weather or waterproofed walls, floors, and roofs.
- B. Flash vent and soil pipes with 700mm by 700 mm sheet lead, minimum 200 mm above roof. Counterflash with caulked stack flashing sleeve, Zurn Z-196 or equal.
- C. Flash floor drains over finished areas and roof drains with lead, 254 mm clear on sides, minimum 915 mm x 915 mm sheet size. Fasten flashing to drain with clamping device.
- D. Provide 200 mm minimum height curbs for roof-mounted mechanical equipment. Flash and counter flash with galvanized steel, soldered and waterproofed.
- E. Provide continuous lead or neoprene safes below air supply unit casing, built-up mop sinks, shower stalls, shower room floors located above rooms. Solder at joints, flash into floor drains, and turn up 150 mm into walls or to top of curbs, and caulk into joints.

3.6 SLEEVES

- A. Set sleeves in position in advance of concrete work. Provide suitable reinforcing around sleeves.
- B. Where piping or ductwork passes through floor, ceiling or wall, close off space between pipe or duct and construction with noncombustible insulation. Provide tight-fitting metal caps on both sides and caulk.
- C. Install chrome-plated escutcheons where piping passes through finished surfaces.
- D. Provide pipe sleeves for all mechanical piping.
- E. Adequately sleeve pipe passing through concrete or masonry walls or concrete slabs to receive both pipe and insulation pertaining thereto.

- F. Waterproof sleeves shall be Thunderline Link-Seal, Calpico Sealing Linx or equal.
- G. Seal pipes passing through walls or slabs. Use mastic or oakum seal in the annular space in nonfire-rated walls.
- H. Insulated pipe shall be insulated in sleeves, sealed, and pointed as above.
- I. Install sleeves on pipes as they are being hung, ready for proper placement in wall as wall is being constructed.
- J. Where sleeves have been inadvertently omitted in concrete floors, provide requisite pipe opening by using properly sized diamond core drills. Areas located below drilling operations shall be protected from possible damage.

3.7 SEISMIC EXPANSION COMPENSATORS

- A. Provide seismic expansion compensators where piping crosses Structural Isolation Breaks (SIBs) between buildings. Refer to Section 15245 Mechanical Seismic Control for amount of compensation required.
- B. Provide flexible duct connections at SIBs. Refer to Section 15240, article "Flexible Duct Connections". Refer to Section 15245 Mechanical Seismic Control for amount of compensation required.

- END -